

Exercise 0:

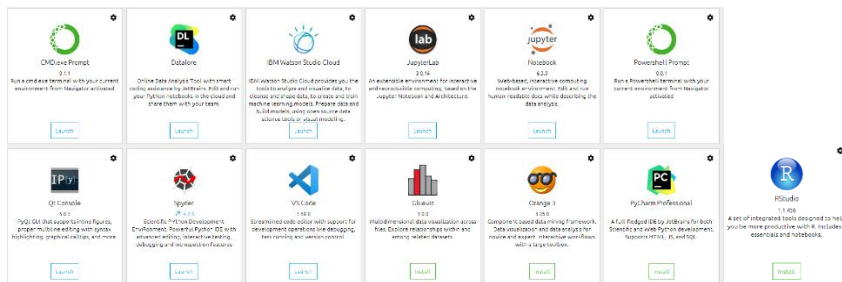
python version Visit <https://www.anaconda.com/>. What is the current version of Python in Anaconda available on the website?

Answer: Python 3.8

**Exercise 2:**

check the anaconda navigator Now, you have to check whether Anaconda is installed in your machine. Click “start” and look for the Anaconda folder, and click “Anaconda Navigator”. At the Anaconda Navigator, click at the “Home” tab. How many applications available in Anaconda? List all of them here.

Answer:



On left hand side of Navigator Click the “Environments” tab, how many packages are installed in the Anaconda package installed in your machine?

Answer:

☒ astroid	
☒ astropy	
☒ async-generator	

362 packages available

How many packages are available in the Anaconda? Those are ready for you to install.

☒ astroid	
☒ astropy	
☒ async-generator	

Answer: 365 packages available

Exercise 3: play with anaconda console

```
`conda list anaconda`
```

What is the output? and what is the current version of Anaconda that is installed in your machine?

Answer:

```

C:\Users\phuri>conda list anaconda
# packages in environment at C:\Users\phuri\Miniconda3:
#
# Name                      Version                      Build      Channel
_anaconda_depends           2020.07                      py38_0
anaconda                    custom                       py38_1
anaconda-client             1.8.0                        py38haa95532_0
anaconda-project            0.10.1                       pyhd3eb1b0_0

```

Next, you need to check the installed packages by using the following command

```
`conda list`
```

Is the list that appear in the command line mode same with the list of installed packages that you saw in the Anaconda navigation?

Answer: long list of packages, that can take several pages to capture all of them.

```

# ❌ - conda list
# packages in environment at C:\Users\phuri\Miniconda3:
#
# Name                                Version                                Build                                Channel
_anaconda_depends                     2020.07                               py38_0
alabaster                              0.7.12                                pyhd3eb1b0_0
anaconda                               custom                                py38_1
anaconda-client                        1.8.0                                 py38haa95532_0
anaconda-project                       0.10.1                                pyhd3eb1b0_0
aniso8601                              9.0.1                                  pypi_0
anyio                                  2.2.0                                 py38haa95532_2
appdirs                               1.4.4                                 py_0
argh                                   0.26.2                                py38_0
argon2-cffi                           20.1.0                                py38h2bbff1b_1
args                                  0.1.0                                 pypi_0
asn1crypto                            1.4.0                                 py_0
astroird                              2.6.6                                 py38haa95532_0
astropy                               4.3.0.post1                           py38h08aeedc_1
async_generator                       1.10                                  pyhd3eb1b0_0
atomicwrites                          1.4.0                                 py_0
attrs                                 21.2.0                                pyhd3eb1b0_0
autopep8                              1.5.6                                 pyhd3eb1b0_0
babel                                 2.9.1                                 pyhd3eb1b0_0
backcall                              0.2.0                                 pyhd3eb1b0_0
backports                             1.0                                   pyhd3eb1b0_2
backports.shutil_get_terminal_size    1.0.0                                 pyhd3eb1b0_3
bcrypt                                3.2.0                                 py38he77u522_0
beautifulsoup4                         4.9.3                                 pyha847dfe_0
bitarray                              2.3.0                                 py38h2bbff1b_1
bkcharts                              0.2                                   py38_0
black                                 19.10b0                               py_0
blas                                  1.0                                   mkl
bleach                                4.0.0                                 pyhd3eb1b0_0
blosc                                 1.21.0                                h19ae4d4_0
bokeh                                 2.3.3                                 py38haa95532_0
boto                                  2.49.0                                py38_0
bottleneck                            1.3.2                                 py38h2a96729_1
brotli                               1.0.9                                 ha925a31_2
brltyip                               0.7.0                                 py38h2bbff1b_1093
bz2p2                                 1.0.8                                 he77u522_0
ca-certificates                       2021.7.5                              haa95532_1
certifi                               2021.5.30                             py38haa95532_0
cffi                                  1.14.6                                py38h2bbff1b_0
cfitsio                               3.470                                 he77u522_6
chardet                               4.0.0                                 py38haa95532_1093
charls                                2.2.0                                 h6c2663c_0
click                                 8.0.1                                 pyhd3eb1b0_0
clint                                 0.5.1                                  pypi_0
cloudpickle                            1.6.0                                 py_0
clyent                                1.2.2                                 py38_1

```

Now, you need to learn how to update Anaconda. First, it is better to update the package manager (conda). Type

```
`conda update conda`
```

Are there any packages need to be updated? What are they?

Answer:

```
cffi 1.14.3-py38hcd4344a_2 → 1.14.6-py38h2bbff1b_0
chardet 3.0.4-py38haa95532_1003 → 4.0.0-py38haa95532_1003
conda-package-handling 1.7.2-py38h76e460a_0 → 1.7.3-py38h8cc25b3_1
cryptography 3.2.1-py38hcd4344a_1 → 3.4.7-py38h71e12ea_0
importlib_metadata pkgs/main/win-64::importlib_metadata~ → pkgs/main/noarch::importlib_metadata-3.10.0-hd3eb1b0_0
keyring 22.3.0-py38haa95532_0 → 23.0.1-py38haa95532_0
lz4-c 1.9.3-h2bbff1b_0 → 1.9.3-h2bbff1b_1
numba 0.53.0-py38hf11a4ad_0 → 0.53.1-py38hf11a4ad_0
pandocfilters conda-forge/noarch::pandocfilters-1.4~ → pkgs/main/win-64::pandocfilters-1.4.3-py38haa95532_1
pip 20.2.4-py38haa95532_0 → 21.0.1-py38haa95532_0
pyopenssl 19.1.0-pyhd3eb1b0_1 → 20.0.1-pyhd3eb1b0_1
python-dateutil 2.8.1-pyhd3eb1b0_0 → 2.8.2-pyhd3eb1b0_0
pywin32 227-py38he774522_1 → 228-py38hbaba5e8_1
pyzmq 20.0.0-py38hd77b12b_1 → 22.2.1-py38hd77b12b_1
requests 2.24.0-py_0 → 2.25.1-pyhd3eb1b0_0
ruamel_yaml 0.15.87-py38he774522_1 → 0.15.100-py38h2bbff1b_0
setuptools 50.3.1-py38haa95532_1 → 52.0.0-py38haa95532_0
six pkgs/main/win-64::six-1.15.0-py38haa9~ → pkgs/main/noarch::six-1.16.0-pyhd3eb1b0_0
sqlite 3.33.0-h2a8f88b_0 → 3.36.0-h2bbff1b_0
tqdm 4.51.0-pyhd3eb1b0_0 → 4.62.0-pyhd3eb1b0_1
urllib3 1.25.11-py_0 → 1.26.6-pyhd3eb1b0_1
vc 14.1-h0510ff6_4 → 14.2-h21ff451_1
vs2015_runtime 14.16.27012-hf0eaf9b_3 → 14.27.29016-h5e58377_2
wheel 0.35.1-pyhd3eb1b0_0 → 0.37.0-pyhd3eb1b0_0
zeromq 4.3.3-ha925a31_3 → 4.3.4-hd77b12b_0
```

Exercise 4: install a new package using anaconda prompt You need to know how to install new packages using Anaconda prompt in cases that the installed packages are not enough to solve your problem. We use Git package as an example. Git is one of the popular version control package that we will try it later.

Type

```
`conda update git`
```

What is the output?

Answer:

```
🐍 ❌ ~ conda update git

PackageNotInstalledError: Package is not installed in prefix.
  prefix: C:\Users\phuri\Miniconda3
  package name: git
```

Type

`conda install git`

Answer:

```

❗ ~ conda install git
Collecting package metadata (current_repodata.json): done
Solving environment: done

# Package Plan #

environment location: C:\Users\phuri\Miniconda3

added / updated specs:
- git

The following packages will be downloaded:

package | build | size
-----|-----|-----
git-2.32.0 | haa95532_1 | 50.9 MB
Total: 50.9 MB

The following NEW packages will be INSTALLED:

git | phgs/main/win-64::git-2.32.0-haa95532_1

Proceed ([y]/n)?

Downloading and Extracting Packages
git-2.32.0 | 50.9 MB | ##### | 100%
Preparing transaction: done
Verifying transaction: done
Executing transaction: | DEBUG menuinst_win32: _init_(198): Menu: name: 'Anaconda${PY_VER} ${PLATFORM}', prefix: 'C:\Users\phuri\Miniconda3', env_name: 'None', mode: 'user', used_mode: 'user'
DEBUG menuinst_win32:create(323): Shortcut cmd is C:\Users\phuri\Miniconda3\python.exe, args are ['C:\Users\phuri\Miniconda3\comp.py', 'C:\Users\phuri\Miniconda3', 'C:\Users\phuri\Miniconda3\Library\bin\bash.exe', '--login', '-i', '--']
done

```

Check the version of the Git package by using

`git --version`

command What is the output?

Answer:

```

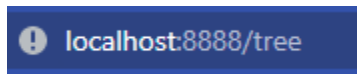
🐍 ❗ ~ git --version
git version 2.32.0.windows.1

```

Exercise 5: play with the jupyter notebook

Let's start, at the Anaconda navigator, click "Jupyter Notebook" (not Jupyterlab). You will see the Jupyter notebook on your browser with the list of files. What is the URL of the Jupyter notebook that shows on your browser?

Answer: localhost:8888



Type print('Hello World') in the first cell and run. The output becomes visible right underneath the cell.

What is the output?

Answer: Hello World

```

In [2]: print("Hello World")

Hello World

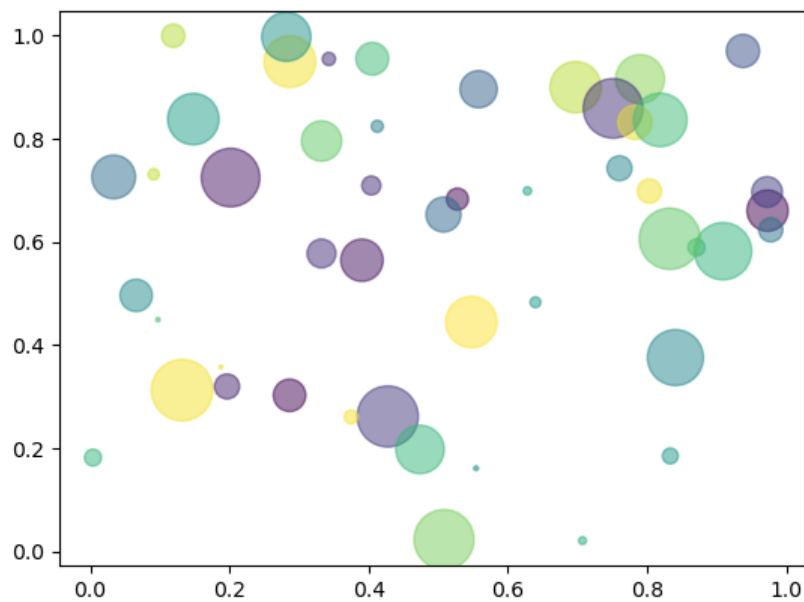
```

Next, we will test how to create a figure on the Jupyter notebook. Type the following code in the second cell and run.

```
import numpy as np
import matplotlib.pyplot as plt
np.random.seed(999)
N = 50
x = np.random.rand(N)
y = np.random.rand(N)
colors = np.random.rand(N)
area = (30 * np.random.rand(N))**2
plt.scatter(x, y, s=area, c=colors, alpha=0.5)
plt.show()
```

Answer: scatter plotsssss

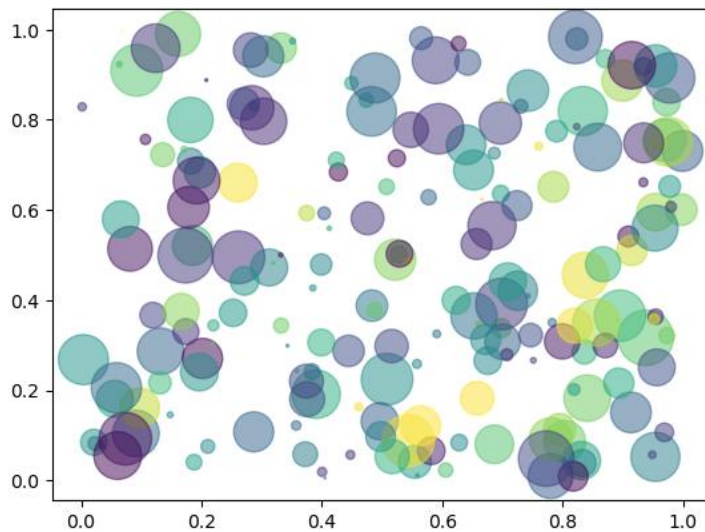
```
In [3]: import numpy as np
import matplotlib.pyplot as plt
np.random.seed(999)
N = 50
x = np.random.rand(N)
y = np.random.rand(N)
colors = np.random.rand(N)
area = (30 * np.random.rand(N))**2
plt.scatter(x, y, s=area, c=colors, alpha=0.5)
plt.show()
```



You have to modify the figure by adding 50 more circles. Please, try to edit the code above. What is your new code that add 50 more circles?

Answer: change \$N\$

```
In [6]: import numpy as np
import matplotlib.pyplot as plt
np.random.seed(999)
N = 200
x = np.random.rand(N)
y = np.random.rand(N)
colors = np.random.rand(N)
area = (30 * np.random.rand(N))**2
plt.scatter(x, y, s=area, c=colors, alpha=0.5)
plt.show()
```



Next, the Jupyter notebook also supports displaying table objects. Type the following code in the next cell and run.

```
import pandas as pd
df = pd.read_csv("https://raw.githubusercontent.com/plotly/datasets/master/school_earnings.csv")
df
```

Answer: read csv file from link provided

```
In [7]: import pandas as pd
df = pd.read_csv("https://raw.githubusercontent.com/plotly/datasets/master/school_earnings.csv")
df
```

Out[7]:

	School	Women	Men	Gap
0	MIT	94	152	58
1	Stanford	96	151	55
2	Harvard	112	165	53
3	U.Penn	92	141	49
4	Princeton	90	137	47
5	Chicago	78	118	40
6	Georgetown	94	131	37
7	Tufts	76	112	36
8	Yale	79	114	35
9	Columbia	86	119	33
10	Duke	93	124	31
11	Dartmouth	84	114	30
12	NYU	67	94	27
13	Notre Dame	73	100	27
14	Cornell	80	107	27
15	Michigan	62	84	22
16	Brown	72	92	20
17	Berkeley	71	88	17
18	Emory	68	82	14
19	UCLA	64	78	14
20	SoCal	72	81	9

In the next cell, type `print(df)` and run. What is the difference between `df` from the previous command and `print(df)`, and why it is different?

Answer:

```
In [8]: print(df)
```

```

      School  Women  Men  Gap
0      MIT      94  152   58
1  Stanford      96  151   55
2   Harvard     112  165   53
3   U.Penn      92  141   49
4  Princeton      90  137   47
5   Chicago      78  118   40
6  Georgetown      94  131   37
7     Tufts      76  112   36
8     Yale      79  114   35
9   Columbia      86  119   33
10     Duke      93  124   31
11  Dartmouth      84  114   30
12     NYU      67   94   27
13  Notre Dame      73  100   27
14   Cornell      80  107   27
15  Michigan      62   84   22
16   Brown      72   92   20
17  Berkeley      71   88   17
18   Emory      68   82   14
19   UCLA      64   78   14
20   SoCal      72   81    9

```

Exercise 6: markdown and code in jupyter notebook

The Jupyter notebook also supports latex's mathematical formula. Type the following code in the next cell, and run.

```
%%latex
\begin{align}
&8 \\
\mathbf{B}' &= -\nabla \times \mathbf{E}, \\
\mathbf{E}' &= \nabla \times \mathbf{B} - 4\pi \mathbf{j}, \\
x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
\end{align}
```

What is the output?

Answer:

```
%%latex
```

$$\begin{aligned} \mathbf{B}' &= -\nabla \times \mathbf{E}, \\ \mathbf{E}' &= \nabla \times \mathbf{B} - 4\pi \mathbf{j}, \\ x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{aligned}$$

Given the limit of the following function

$$\lim_{\chi \rightarrow \infty} \frac{\sin(\chi)}{\chi} = 0$$

Answer:

```
\begin{align}
\lim_{\chi \rightarrow \infty} \frac{\sin(\chi)}{\chi} &= 0 \\
\end{align}
```

$$\lim_{\chi \rightarrow \infty} \frac{\sin(\chi)}{\chi} = 0$$

Create derivative of equation in Jupyter Notebook.

Answer:

```
\begin{align}
\left. \frac{df}{dx} \right|_{x=a} &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \\
\end{align}
```

$$\left. \frac{df}{dx} \right|_{x=a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$